

Trajectory Clustering in Vehicle Telematics Data

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Introduction & objectives

“Our objective is to accurately and efficiently cluster dumpers trajectories over an extended period.”

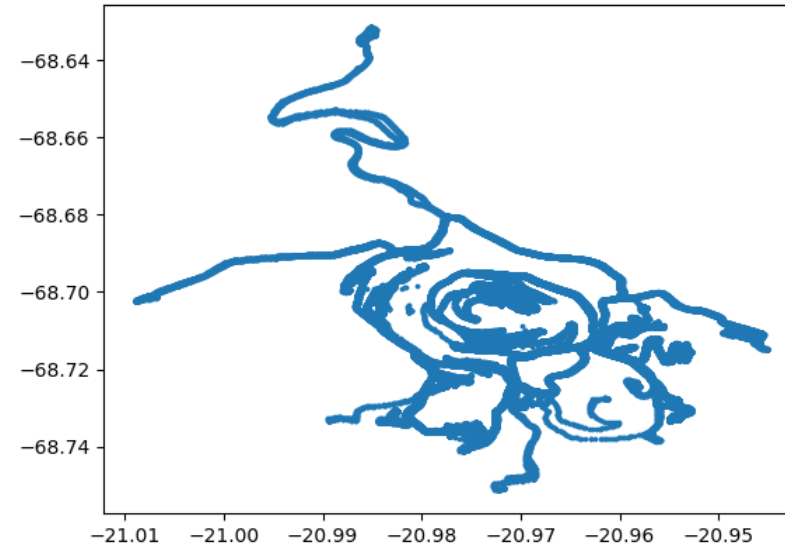


Introduction & objectives

DataSet

Mine	Collahuasi
Vehicle	CA-158
Periode	January 2023
Sensor	MEMS
Frequency	0.2 Hz

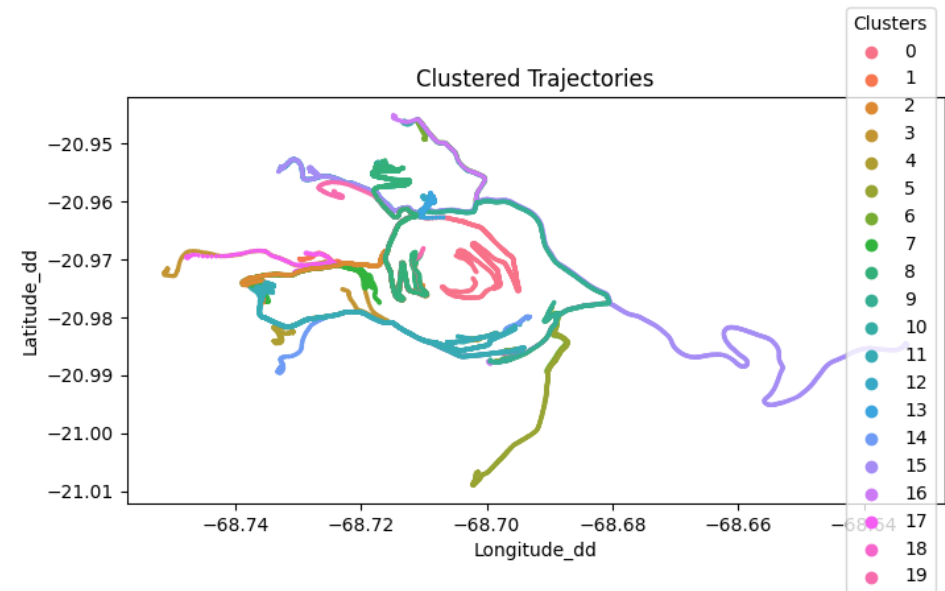
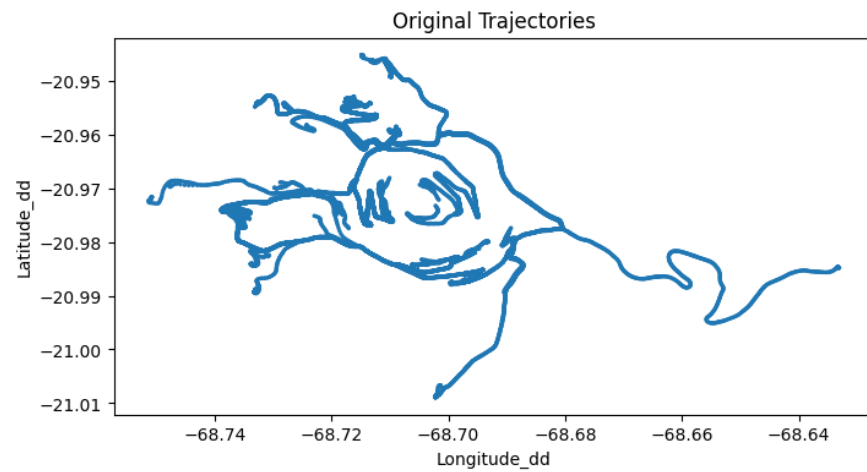
Trajectories



Collahuasi – Tarapacá, Chili

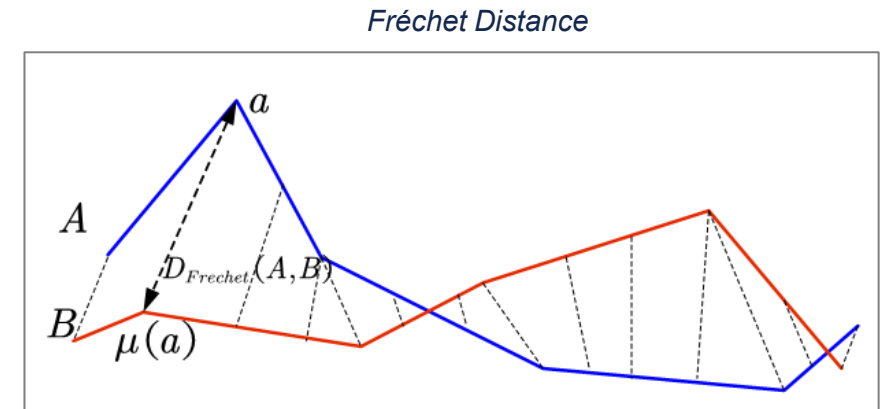
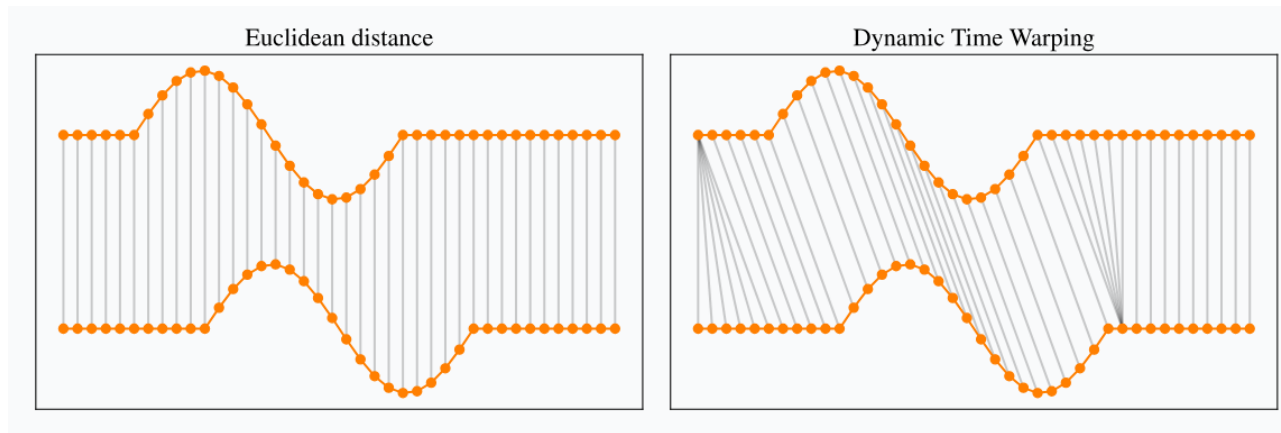


Introduction & objectives



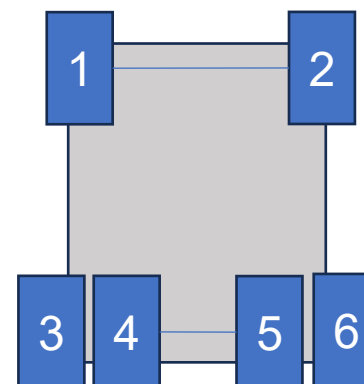
Distance measures in spatial trajectory data clustering

Euclidean Distance Dynamic Time Warping (DTW) Hausdorff Distance Fréchet Distance, ...



Data Description

```
Index(['Time_utc', 'VehicleName', 'VehicleType', 'Pressure_Pa_1',
      'Pressure_Pa_2', 'Pressure_Pa_3', 'Pressure_Pa_4', 'Pressure_Pa_5',
      'Pressure_Pa_6', 'Temperature_K_1', 'Temperature_K_2',
      'Temperature_K_3', 'Temperature_K_4', 'Temperature_K_5',
      'Temperature_K_6', 'ColdPressure_Pa_1', 'ColdPressure_Pa_2',
      'ColdPressure_Pa_3', 'ColdPressure_Pa_4', 'ColdPressure_Pa_5',
      'ColdPressure_Pa_6', 'Latitude_dd', 'Longitude_dd', 'Altitude_m',
      'Speed_mps', 'epsilonX', 'epsilonY', 'epsilonZ', 'epsilonSpeed',
      'AtmosphericPressure_Pa', 'dt_s', 'dd_m', 'RatioPayload'],
      dtype='object')
```



	Time_utc	VehicleName	VehicleType	Pressure_Pa_1	Pressure_Pa_2	Pressure_Pa_3	Pressure_Pa_4	Pressure_Pa_5	Pressure_Pa_6	Temperature_K_1	...
0	2023-01-01 00:00:00.328	C-132	Dumper	993010.933333	1.003995e+06	856000.0	798005.466667	819994.533333	864994.533333	346.15	...
1	2023-01-01 00:00:05.338	C-132	Dumper	993177.933329	1.003911e+06	856000.0	798088.966665	819911.033335	864911.033335	346.15	...
2	2023-01-01 00:00:10.331	C-132	Dumper	993344.366669	1.003828e+06	856000.0	798172.183334	819827.816666	864827.816666	346.15	...
3	2023-01-01 00:00:15.325	C-132	Dumper	993510.833331	1.003745e+06	856000.0	798255.416666	819744.583334	864744.583334	346.15	...
4	2023-01-01 00:00:20.448	C-132	Dumper	993681.600000	1.003659e+06	856000.0	798340.800000	819659.200000	864659.200000	346.15	...

5 rows x 39 columns

Data Description

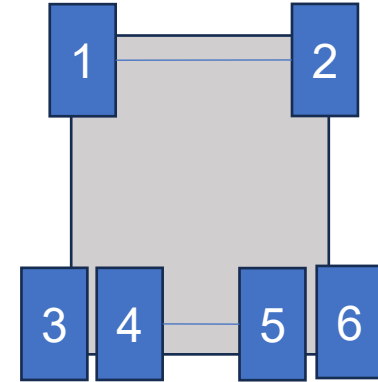
```
Index(['Time_utc', 'VehicleName', 'VehicleType', 'Pressure_Pa_1',
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      'Pressure_Pa_6', 'Temperature_K_1', 'Temperature_K_2',
      'Temperature_K_3', 'Temperature_K_4', 'Temperature_K_5',
      'Temperature_K_6', 'ColdPressure_Pa_1', 'ColdPressure_Pa_2',
      'ColdPressure_Pa_3', 'ColdPressure_Pa_4', 'ColdPressure_Pa_5',
      'ColdPressure_Pa_6', 'Latitude_dd', 'Longitude_dd', 'Altitude_m',
      'Speed_mps', 'epsilonX', 'epsilonY', 'epsilonZ', 'epsilonSpeed',
      'AtmosphericPressure_Pa', 'dt_s', 'dd_m', 'RatioPayload'],
      dtype='object')
```

```
df['RatioPayload']
```

0	0.0
1	0.0
2	0.0
3	0.0
4	0.0
...	...
201307	1.0
201308	1.0
201309	1.0
201310	1.0
201311	1.0

Empty

Loaded



	Time_utc	VehicleName	VehicleType	Pressure_Pa_1	Pressure_Pa_2	Pressure_Pa_3	Pressure_Pa_4	Pressure_Pa_5	Pressure_Pa_6	Temperature_K_1	...
0	2023-01-01 00:00:00.328	C-132	Dumper	993010.933333	1.003995e+06	856000.0	798005.466667	819994.533333	864994.533333	346.15	...
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5 rows x 39 columns

RESOURCES

<https://github.com/mohamedELBAHA/5AIMDS>

 mohamedELBAHA

add starter ipynb

01b754f · last year

🕒 15 Commits

README.md

Update README.md

last year

Requirement.txt

Add requir

last year

dataDumper.parquet

upload data

last year

mines.png

Add files via upload

last year

starter_notebook.ipynb

add starter ipynb

last year

README

Advanced Machine Learning - Final Project

Trajectory Clustering in Vehicle Telematics Data

license MIT python 3.7 | 3.8

Welcome to your capstone project for the Advanced Machine Learning course. This project aims to bridge theory and practice by applying ML techniques to real-world telematics data collected from MEMS (Micro-Electro-Mechanical Systems) sensors in vehicles. You will engage in a comprehensive analysis that includes data preprocessing, exploratory data analysis, distance metrics for trajectories, modeling, and interpretation of results. This project is designed to reinforce your understanding of machine learning concepts and enhance your practical skills.



Advanced Machine Learning - Final Project

Readme

Activity

1 star

1 watching

2 forks

Releases

No releases published

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Packages

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Languages

Jupyter Notebook 100.0%